higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

INDUSTRIAL INSTRUMENTS N5

18 August 2026

This marking guideline consists of 6 pages.

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TVET Exam Papers NATED & NCV	**N1-N6** Mathematics NATED	**N1-N4** Engineering Science NATED	**N1-N6** Industrial Electronics NATED	**N4-N6** Electrotechnics NATED	**N4-N6** Mechanotechnics NATED
N5-N6 Power Machines NATED	**N4-N6** Quantity Surveying NATED	**N5-N6** Strength of Materials & Structures NATED	**N1-N3** Engineering Drawing NATED	**N3** Eletro-technology NATED	**N3** Mechano-technology NATED
N4-N6 Building Administration NATED	**N4-N6** Building& Structural Surveying NATED	**N4-N6** Building & Structural Construction NATED	**N1-N3** Building Science NATED	**N1-N3** Building Drawing NATED	**N4** Mechanical Draughting NATED
N1-N3 Electrical Trade Theory NATED	**N2-N3** Diesel Trade Theory NATED	**N5-N6** Strength of Materials & Structures NATED	**N3** Building & Civil Technology NATED	**N1-N2** Fitting and Machining Theory NATED	**N1-N4** Plating and Structural Steel Drawing NATED
N1-N3 Motor Trade Theory NATED	**N5-N6** Mechanical Drawing & Design NATED	**N1-N2** Plumbing Theory NATED	**N4-N6** Fault Finding & Protective Devices NATED	**N2-N3** Instrument Trade Theory NATED	**N5-N6** Engineering Physics NATED
N5-N6 Fluid Mechanics NATED	**N4-N6** Supervisory Management NATED	**N4-N6** Digital Electronics NATED	**N6** Control Systems NATED	**N1-N2** Bricklaying &Plastering Theory NATED	More…

SECTION A

QUESTION 1

1.1 It is a fluid where the volume and density change✓ with a change in pressure.✓ (2)

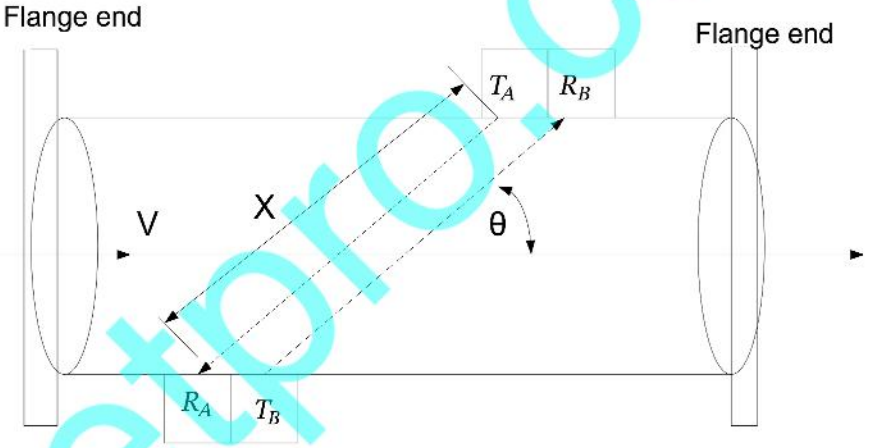
1.2
- Orifice plate
- Venturi tube
- Flow nozzle
- Pitot tube
- Dall tube

(Any 3 × 1) (3)

1.3
1.3.1 C
1.3.2 C **OR** D
1.3.3 C
1.3.4 C
1.3.5 D

(5 × 1) (5)

1.4



✓ (5)

1.5
- Barium titanate
- Lead zirconate titanate

(2)
[17]

QUESTION 2

2.1 Open channel flow measurement instruments are used to monitor the flowrate of liquids✓ through a surface open to the atmosphere.✓ (2)

2.2
2.2.1 rotor
2.2.2 rotate
2.2.3 velocity
2.2.4 meter

(4 × 1) (4)

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2.3 $Q = \frac{2}{3}$ B $\sqrt{2gH^3}$✓

$= \frac{2}{3}$ $(1,2)\sqrt{2 \times 10 \times 1,5^3}$✓✓

=14,789 m³/s✓ (4)

2.4 2.4.1 C
2.4.2 A
2.4.3 D
2.4.4 B

(4 × 1) (4)
[14]

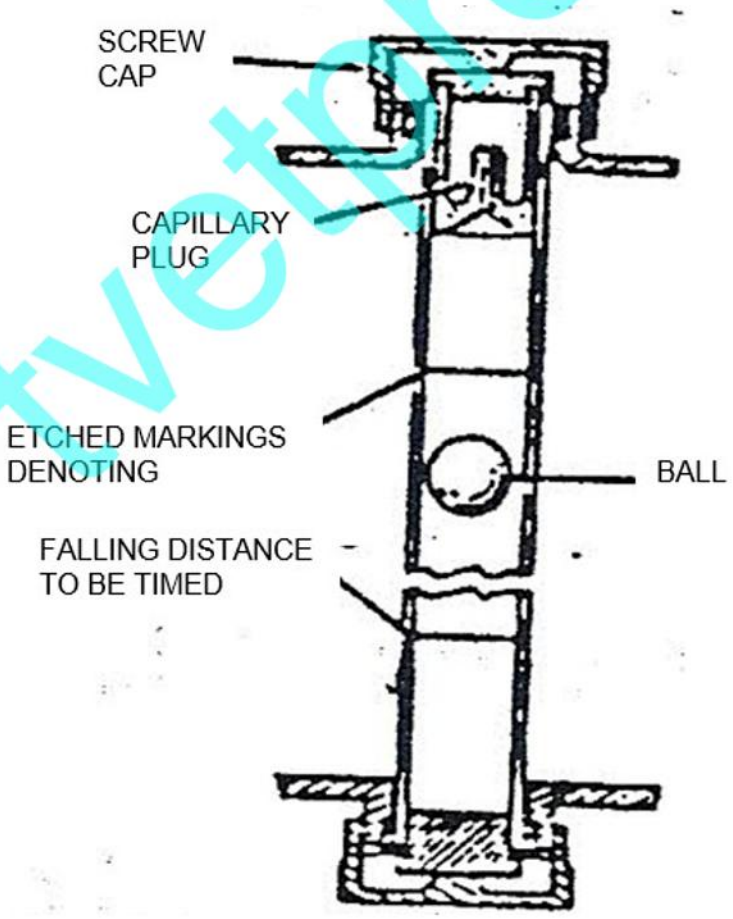
TOTAL SECTION A: 31

SECTION B

QUESTION 3

3.1 Stokes' law relates the viscosity of gases and liquids to the steady-state velocity of a known weight sphere✓ falling through a fluid of known density at very low Reynolds Numbers.✓ (2)

3.2



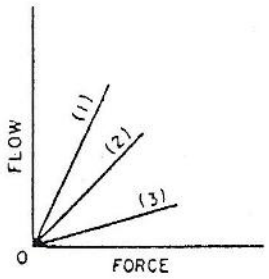
(6)

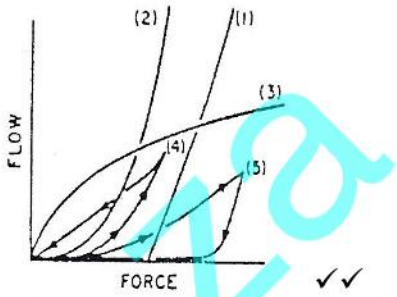
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3.3 Newtonian fluids: When fluids are deformed by strain the ratio between shear rate and shear stress will be a constant.✓✓

Non-Newtonian fluids: When fluids are deformed by strain the ratio between shear rate and shear stress will not be a constant value.✓✓



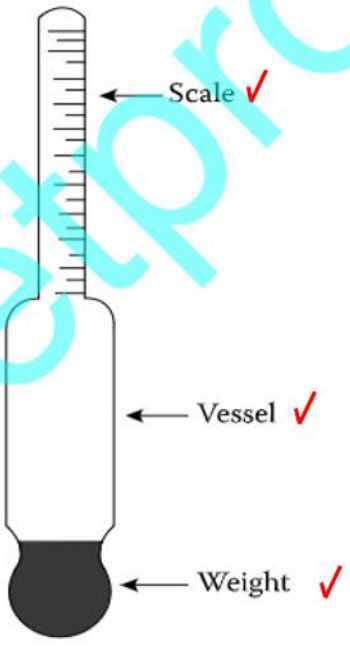


✓✓

(1 mark for each sketch. NB: The students only need to draw one line.) (6)

3.4
- Hair hygrometer
- Wire electrode hygrometer
- Electrolysis type hygrometer
- Resistive type
- Capacitive type
- Microwave (6)

3.5



(3)

3.6 It is the temperature at which the air becomes saturated with water vapour✓ and condensation (like dew) begins to form.✓ (2)

3.7 Poise✓✓ **OR** centipoise✓✓ (2)
[27]

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QUESTION 4

4.1 $pH = -\log[H^+]$✓

$H^+ = 10^{-pH}$✓

$= 10^{-9,79}$

$= 1,622 \times 10^{-10}$ mole/dm^{-5}✓

$pOH = 14 - pH$✓

$= 14 - 9,79$

$= 4,21$✓

$[OH^-] = 10^{-4,21}$✓

$= 6,2 \times 10^{-5}$ mole/dm^{-5} (6)

4.2 By adding a regulated power supply✓ to drive a transformer. ✓ (2)

4.3 A stage of amplification is added to increase sensitivity✓ and reduce non□linearity.✓ (2)

4.4
- It must have resistance to corrosion
- It must have strength against thermal shock
- It must be impermeable to water vapour at high temperature
- It must have strength to mechanical shock (4)

[14]

QUESTION 5

5.1 5.1.1 Gain is the output divided by the input.

5.1.2 It is when the calibration of an instrument gradually shifts over a period of time.

5.1.3 It is defined as the action of the controller, the output signal of which changes at a rate that is proportional to the measured deviation changes (rate).

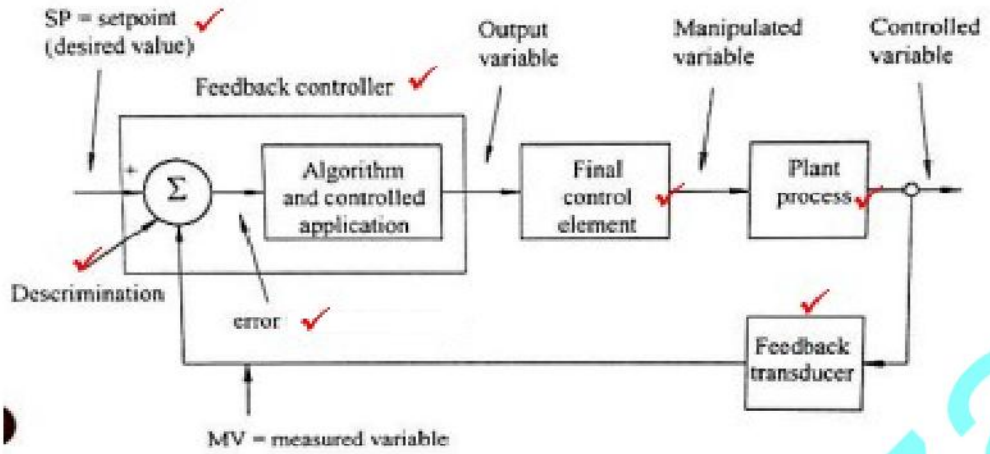
5.1.4 Can be defined as the action of a controller, the output of which is proportional to the rate at which the measured deviation changes (rate).

(4 × 2) (8)

5.2 The integral action will integrate any differences between the measurement and set point,✓ until this difference is zero.✓ (2)

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5.3



(7)

5.4
- Pneumatic signal
- Electric signal
- Hydraulic signal

(3)

5.5 Failure to eliminate offset (2)

5.6
- Differential amplifier or comparator
- Wheatstone bridge
- Linear variable differential transformer
- Linear variable capacitive transducer

(4)

5.7
- Continuous bleed type
- Non-bleed type

(2)
[28]

TOTAL SECTION B: 69
GRAND TOTAL: 100

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